Amir Beck, Chapter 1 - Exercise 1.7

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1.7. Suppose that $\mathbf{R}^n$ and $\mathbf{R}^m$ are equipped with norms $||.||_b$ and $||.||_a$, respectively, and let $\mathbf{A} \in \mathbb{R}^{m \times n}$.
Show that there exists $\mathbf{x} \in \mathbb{R}^n$ such that $||\mathbf{x}||_a \leq 1$ and $||\mathbf{Ax}||_b = ||\mathbf{A}||_{a,b}$.

Recall that, by definition,

$$||\mathbf{A}||_{a,b} := \max_{||\mathbf{x}||_a \leq 1} ||\mathbf{Ax}||_b$$

We showed, in exercise 1.6, that for every norm $||.||$ on $\mathbb{R}^n$, the function $\mathbf{x} \mapsto ||\mathbf{x}||$ is continuous. Therefore, since the function $\mathbf{x} \mapsto \mathbf{Ax}$ is also continuous, we have that the function

$$\mathbf{x} \mapsto ||\mathbf{Ax}||_b$$

is continuous.

The set

$$\{||\mathbf{x}||_a \leq 1\} := \{\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}||_a \leq 1\}$$

is closed and bounded, and therefore compact. Applying Weierstrass theorem to the function $f(\mathbf{x}) = ||\mathbf{Ax}||_b$ over the set $\{||\mathbf{x}||_a \leq 1\}$ we can conclude that f attains a maximum over this set. That is, there exists $\mathbf{x_0} \in \mathbb{R}^n$, such that $||\mathbf{x}||_a \leq 1$ and

$$||\mathbf{A}||_{a,b} = ||\mathbf{Ax_0}||_b.$$